

Complete React Notes (Beginner to Advanced) — FULL THEORY + PRACTICAL

1. What is React? (Theory)

React is a **JavaScript Library** used to build **interactive User Interfaces (UI)**.

 **In Simple Words:**

 React helps you build websites like: - Amazon - Flipkart - Instagram

 **Core Idea:**

React breaks UI into **small reusable components**.

2. Why React? (Theory)

Problems in Traditional JS:

- Manual DOM updates
- Code becomes complex
- Hard to maintain

React Solution:

- Uses **Virtual DOM** (fast updates)
 - Component-based architecture
 - Easy to manage large apps
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3. Virtual DOM (Important Interview)

 **Theory:**

Virtual DOM is a **copy of real DOM**.

 **Working:**

1. React creates Virtual DOM

2. Detects changes
3. Updates only changed parts in real DOM

Benefit:

👉 Faster performance

4. Components (Core Concept)

Theory:

Components are **building blocks of UI**.

Types:

1. Functional Components (Modern)
 2. Class Components (Old)
-

Example:

```
function Welcome() {  
  return <h1>Hello Gopi</h1>;  
}
```

Output:

👉 Hello Gopi

5. JSX (JavaScript XML)

Theory:

JSX allows writing HTML inside JavaScript.

```
const element = <h1>Hello React</h1>;
```

6. Props (Data Passing)

Theory:

Props are used to **pass data from parent → child**.

```
function User({ name }) {  
  return <h2>{name}</h2>;  
}  
  
<User name="Gopi" />
```

Output:

👉 Gopi

7. State (Important)

Theory:

State is used to **store dynamic data**.

useState Hook

```
import { useState } from "react";  
  
function Counter() {  
  const [count, setCount] = useState(0);  
  
  return (  
    <div>  
      <h2>{count}</h2>  
      <button onClick={() => setCount(count + 1)}>Increase</button>  
    </div>  
  );  
}
```

Output:

👉 0 → 1 → 2 → 3

8. useEffect Hook

Theory:

Used for **side effects** like: - API calls - Data fetching - DOM updates

```
useEffect(() => {  
  console.log("Component Loaded");  
}, []);
```

9. Event Handling

```
<button onClick={() => alert("Clicked")}>Click</button>
```

10. Conditional Rendering

```
{isLoggedIn ? <h1>Welcome</h1> : <h1>Please Login</h1>}
```

11. Lists & Keys

```
const users = ["A", "B", "C"];  
  
users.map((u, i) => <li key={i}>{u}</li>);
```

12. Forms (Controlled Components)

Theory:

React controls form data using state.

```
const [name, setName] = useState("");  
  
<input value={name} onChange={(e) => setName(e.target.value)} />
```

13. React Router (Navigation)

Theory:

Used for page navigation without reload.

```
<BrowserRouter>  
  <Routes>  
    <Route path="/" element={<Home />} />  
  </Routes>  
</BrowserRouter>
```

14. useNavigate

```
const navigate = useNavigate();  
navigate("/home");
```

15. API Calls (Axios)

Theory:

Used to communicate with backend.

```
useEffect(() => {  
  axios.get("/api/products")  
    .then(res => console.log(res.data));  
}, []);
```

16. useRef

Theory:

Access DOM directly.

```
const ref = useRef();
```

17. useMemo

Theory:

Optimizes performance by caching values.

18. Props Drilling

Problem:

Passing data multiple levels

Solution:

👉 Context API

19. Context API

```
const UserContext = createContext();
```

20. Custom Hooks

Reusable logic.



21. Performance Optimization

- React.memo
- useMemo
- useCallback



REAL-TIME PROJECT: E-COMMERCE APP



Architecture (Theory)

Frontend (React) ↓ REST API Calls Backend (Spring Boot) ↓ Database (MySQL)



Folder Structure

```
src/  
├ components/  
├ pages/  
├ services/  
├ hooks/  
└ App.js
```



Features (Real World)

- Login System
- Product Listing
- Add to Cart
- Order Placement



Example: Product Component

```
function Product({ product }) {  
  return (  
    <div>
```

```
    <h3>{product.name}</h3>
    <p>{product.price}</p>
  </div>
);
}
```

API Layer

```
axios.get("/products")
```

Output

- Dynamic product list
- Navigation between pages
- Cart updates in real-time

INTERVIEW QUESTIONS (WITH THEORY)

1. What is React?

React is a JavaScript library used to build UI using components.

2. What is Virtual DOM?

Virtual DOM is a lightweight copy of real DOM that improves performance.

3. Difference between State and Props?

State	Props
Mutable	Immutable
Internal	External

4. What are Hooks?

Hooks allow using state and lifecycle features in functional components.

5. What is useEffect?

Used for side effects like API calls.

6. Controlled vs Uncontrolled Components?

Controlled → React controls data Uncontrolled → DOM controls data

FINAL SUMMARY

You learned:

✅ React from ZERO ✅ Theory + Practical ✅ Real-time architecture ✅ Interview preparation

NEXT LEVEL (Tell me if needed)

- Redux Toolkit (Advanced State)
 - JWT Authentication
 - Full Stack Project (React + Spring Boot)
 - Deployment
-

💡 Since you said you are a beginner: 👉 Practice each topic step by step 👉 Start with small apps (Counter, Todo) 👉 Then move to full project

🚀 I can also create: - Step-by-step learning plan (Day-wise) - Full project code (frontend + backend) - PDF notes for download